



Michigan Univ connects vehicles

A total of 3,000 vehicles in Ann Arbor, Michigan are taking part in a one-of-a-kind year-long smart car safety pilot project

From the Labs

Applications

Plastic Solar Cells

Researchers at Georgia Tech's Centre for Organic Photonics and Electronics led by Professor Bernard Kippelen, have created the world's first plastic solar cell. By spreading an ultrathin layer of a polymer on the surface of a conducting material produces a strong surface dipole. This converts the air-stable conductors into efficient electrodes. Conductors such as calcium, lithium or magnesium are extremely reactive and will stop working when exposed to oxygen or moisture. This is the reason why solar cells are covered with glass or expensive encapsulating materials. With the plastic solar cell, the conductor can be made of plastic material and will not require any encapsulation. Because of its thin size and ability to bend, you can install these plastic solar cells anywhere – window slides, over cars, on your backpack for all you care.

Packaging

Researchers at Philips have made plastic electronics based RFID tags which resemble a postage stamp. These tags are capable of transmitting multi-bit digital identification codes at 13.56 MHz which is the dominant industry standard for RFID tag applications. Belgian nanotech research group Imec, along with its partners have developed a plastic electronics-based RFID tag with a reader-talks-first communication protocol. In a shopping



Photo Courtesy: Philips

Novatel's Linternity Victory OLED Lamp

mall if many products have such tags, the communicating with the RFID reader will be confusing. According to Imec director Paul Heremans, with the reader-talks-first technology, when the RFID reader powers and contacts the RFID tag on the product, it sends clock and identification data.

OLED lighting

OLED is already used in cellphone screens and in the near future we are going to see a lot of OLED TVs in the market. LG and Samsung already showcased their OLED TVs at CES this year. Germany headquartered Novaled, a leading firm in OLED supply chain has come out with a brand called Linternity to market its OLED products for home use. Check out their table lamp Victory shown alongside which has OLED strips embedded in a wonderfully designed body. It will also save you a log of money on your electricity bills. Now wouldn't you want one of those? There are still other applications whereby light sensitive plastics can be used as sensors, flexible materials can make for wearable materials, interactive packaging using thin film OLED screens, and so on.

The Future?

Will we see plastic electronics replacing silicon electronics? "Plastic electronics is complementary to silicon and is unlikely to replace silicon in applications which are suited to silicon, such as computation," asserts Prof. Friend. The future should accommodate fast silicon electronics with cheaper plastic electronics. 



Bernard Kippelen with the worlds first plastic solar cell



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